Abdulqader Alshihabi

Software Engineer | Backend & AI Engineering

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SUMMARY

Computer Science student graduating in January 2027, with practical experience developing scalable backend systems, full-stack applications, and AI-powered data processing solutions. Experienced in Python, Django, React, TypeScript, and PostgreSQL, with hands-on contributions across enterprise ERP workflows, semantic vector search, and academic research intelligence platforms.

TECHNICAL SKILLS

Programming Languages: Python, TypeScript, JavaScript, SQL, Java, Go

Backend & Cloud: Django, Django REST Framework, RESTful APIs, Role-Based Access Control (RBAC), Supabase, Docker, Azure

Frontend Development: React, Next.js, TypeScript, HTML5, CSS3, Tailwind CSS

Databases & Vector Data: PostgreSQL, pgvector, PL/pgSQL, Relational Schema Design, Query Optimization, Vector Embeddings

AI & Machine Learning: PyTorch, Sentence-Transformers, Vector Search, LLM Pipelines (Ollama/Qwen), Scikit-learn, NLP, Prompt Engineering

Core Methodologies & Tools: Git, GitHub Actions (CI/CD), Unit Testing, SDLC, Agile/Scrum, Linux, Postman, WebSockets

WORK EXPERIENCE

Research & Innovation Consultant (Part-time) **Jul 2026 – Present**

Institute of Research and Consulting Studies, Islamic University of Madinah | Madinah, Saudi Arabia

- Engineered a locally hosted LLM extraction pipeline using Python, Ollama, and Qwen, processing 25,000+ unstructured academic research papers from the university repository, OpenAlex, and Scopus with zero cloud API costs.
- Accelerated comprehensive research data extraction from an estimated several months of manual review to just 10 days, implementing data validation, Arabic text normalization, and retry mechanisms to improve extraction reliability.
- Organized researcher profiles and domain expertise by generating dense vector representations with multilingual Sentence-Transformers and clustering techniques, designing optimized PostgreSQL schemas for relational and vector queries.
- Collaborated with the platform engineer to deliver research analytics for the commercialization engine and prepared technical progress reports and milestone presentations for university leadership.

Software Engineer Intern **Jun 2026 – Jul 2026**

Asaqefah for Business Development | Saaed HR & Payroll ERP | Madinah, Saudi Arabia

- Contributed to Saaed, an enterprise HR and Payroll ERP serving 30+ employees, developed to replace disconnected departmental spreadsheets with integrated transactional models.
- Developed core backend services with Django REST Framework and interactive frontend views with React and TypeScript on a PostgreSQL database.
- Accelerated payroll approval and finance reconciliation workflows by over 4x by automating the transfer of validated payroll calculations directly into Finance general ledgers.
- Implemented role-based access control (RBAC) and strict server-side validation rules to secure sensitive employee records and maintain transactional consistency across cross-departmental workflows.

PROJECTS

Islamic University CRIS – Research Information System | github.com/Abdulqader-Alshihabi/Islamic-University-CRIS

Technologies: React, TypeScript, PostgreSQL, PL/pgSQL, Supabase, Scopus API, OpenAlex API

- Centralized and reconciled 7,000+ academic publications and 70,000+ verified citations across Scopus and OpenAlex APIs, resolving fragmented institutional research records.
- Designed automated ETL pipelines in PostgreSQL and PL/pgSQL with multi-field deduplication algorithms, improving the speed and accuracy of faculty post-publication award evaluations by eliminating manual data entry.
- Developed interactive analytical dashboards and administrative review workflows for institutional research tracking, quartile analytics, and publication verification.

res-handler – AI Semantic Search & Research Matcher | github.com/Abdulqader-Alshihabi/res-handler

Technologies: Python, Go, PostgreSQL, pgvector, Azure OpenAI Embeddings, Docker

- Engineered an AI semantic vector search service that converts unstructured industry problem queries into 1,536-dimensional embedding vectors, indexing 5,000+ academic papers to match faculty expertise.
- Implemented cosine distance similarity lookup using pgvector over dense vector representations, achieving sub-200ms query retrieval latency across high-dimensional embeddings.
- Containerized multi-service deployment with Docker and automated database migrations, ensuring 99.9% production uptime on Azure Container Apps.

Shakkel Learn – Corporate Learning Management System | learn.shakkel.store

Technologies: React, TypeScript, Supabase, Tailwind CSS, Tiptap Editor

- Engineered a multi-tenant corporate learning platform supporting 4 distinct user roles (Administrator, Instructor, Trainee, Corporate Client) with bilingual Arabic/English interfaces.
- Implemented rich interactive course authoring with Tiptap editor, automating 100% of timed assessment evaluations, verified PDF certificates, and HR completion tracking.
- Configured 15+ PostgreSQL Row-Level Security (RLS) policies to enforce tenant-level data isolation and role-based access across corporate learning environments.

EDUCATION

Bachelor of Science in Computer Science **Expected Graduation: January 2027**

Islamic University of Madinah | Madinah, Saudi Arabia

Relevant Coursework: Data Structures & Algorithms, Database Systems, Software Engineering, Artificial Intelligence, Machine Learning, Deep Learning, Operating Systems, Computer Networks

CERTIFICATIONS

- **Advanced Artificial Intelligence (Computer Vision) –** KAUST Academy & Universidad Politécnica de Madrid (2026)
- **Introduction to Artificial Intelligence (with Distinction) –** KAUST Academy & Universidad Politécnica de Madrid (2026)
- **KAUST Academy AI Specialization Selection –** KAUST Academy & Islamic University of Madinah (2026)
- **Asaqefah SandBox Innovation Program –** Asaqefah & Amazon Web Services (AWS) (2026)
- **Introductory Artificial Intelligence Program –** KAUST Academy & NTDP (2025)
- **Improving Deep Neural Networks: Hyperparameter Tuning, Regularization & Optimization –** DeepLearning.AI (2024)
- **Mathematics for Machine Learning: Calculus, Linear Algebra, Probability & Statistics –** DeepLearning.AI (2024)